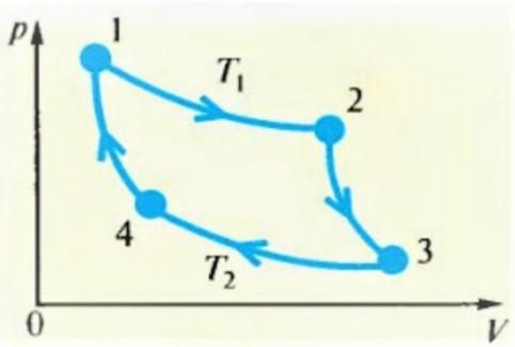
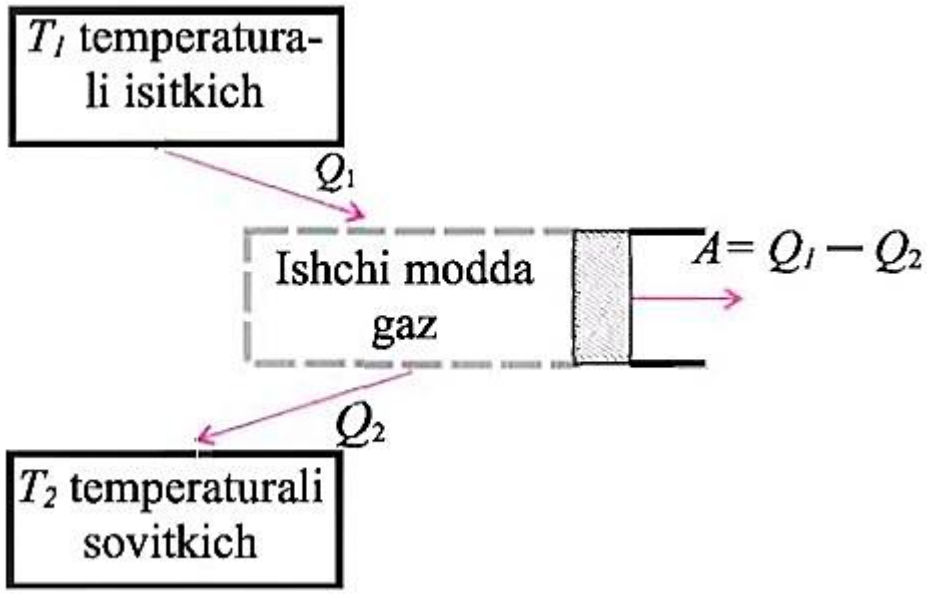
Fizika

9-sinf

Mavzu: Issiqlik dvigatellarining ishlash prinsipi





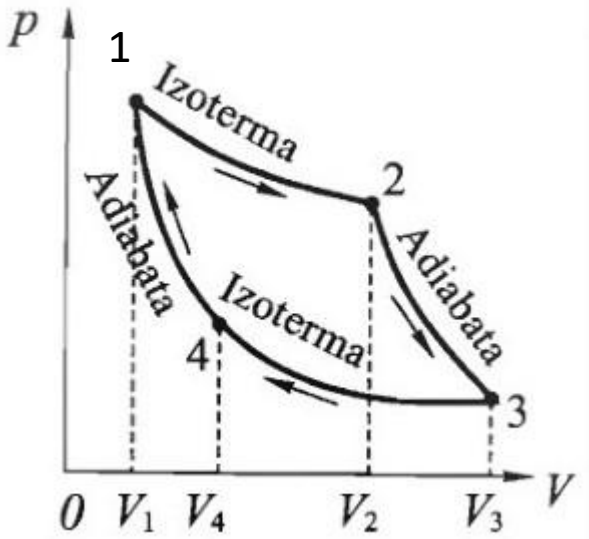
Issiqlik dvigatelining ishlash prinsipi quyidagicha:

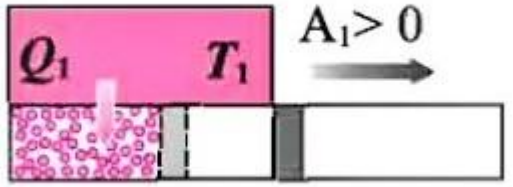
1. Ichki energiyani mexanik energiyaga aylantiradi.

2. Issiqlik dvigatellarining ishlashi uchun turli temperaturali isitkich va sovitkichning bo‘lishi shart.

3. Istalgan issiqlik dvigatelining ishlashi ishchi jism holati o‘zgarishining takrorlanuvchi sikllaridan iborat bo‘ladi.

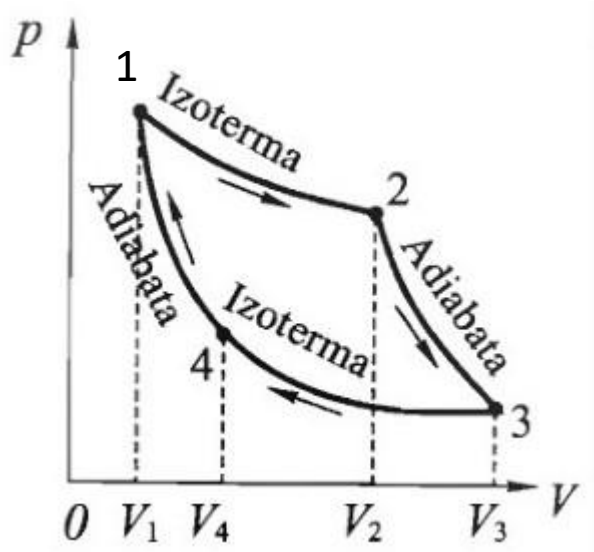
1 - holatda

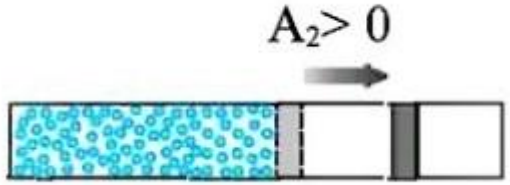




1-holatda turgan gaz T_1 temperaturada izotermik kengayib, 2-holatga o‘tadi. Bu paytda gaz isitkichdan Q_1ga teng issiqlik miqdori olib, tashqi kuchga qarshi A_1 ish bajaradi.

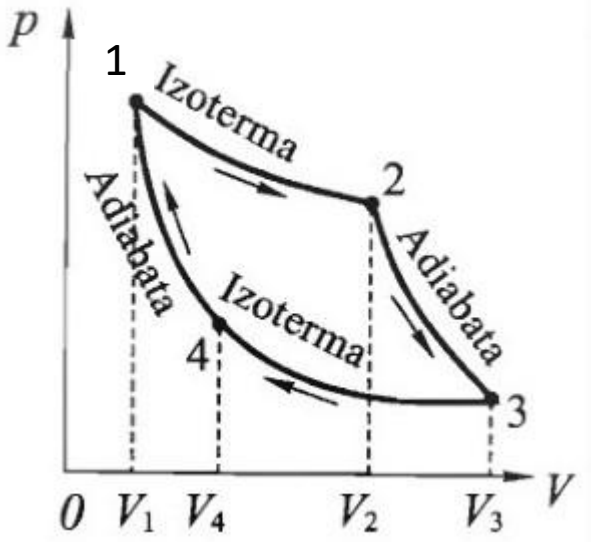
2 - holatda

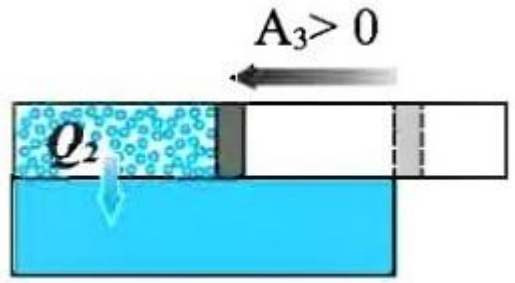




2-holatda gaz isitkichdan ajraladi va adiabatik kengayib, 3-holatga o‘tadi. Bunda gaz o‘zining ichki energiyasi hisobiga tashqi kuchlarga qarshi A_2 ish bajaradi. Gazning ichki energiyasi kamayadi va temperaturasi T_1 dan T_2 gacha pasayadi.

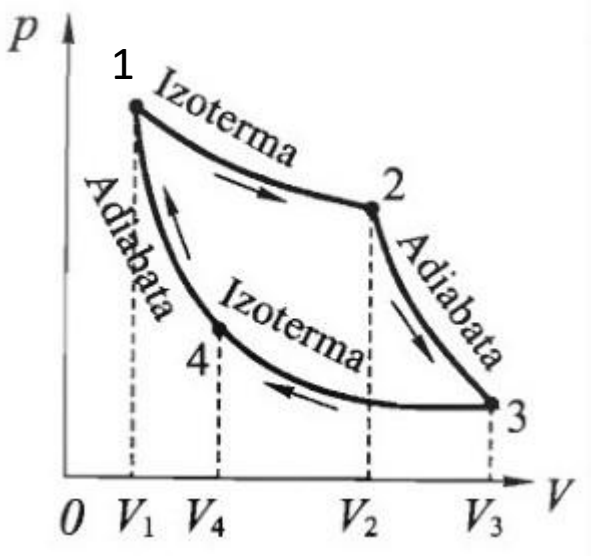
3 - holatda

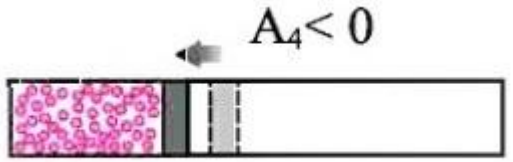




3-holatda gaz sovitgich bilan kontaktga keladi. Gaz 4-holatga o‘tish jarayonida izotermik siqiladi. Bunda tashqi kuchlar gazni siqib A_3 ish bajaradi va ishchi modda sovitgichga Q_2 issiqlik beradi.

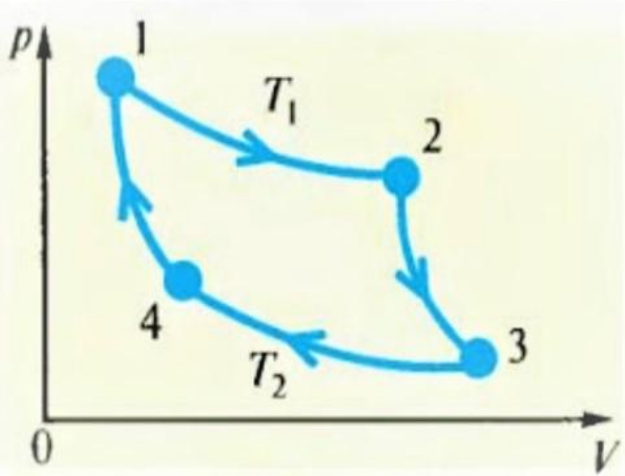
4 - holatda





4-holatda ishchi modda sovitgichdan ajraladi. Bunda gaz adiabatik siqilib 1-holatga qaytadi. Tashqi kuchlar gazga qarshi A_4 ish bajaradi va gazning temperaturasi T_2 dan T_1 gacha ko‘tariladi.

Karno sikli



Karno sikli bo‘yicha ishlayotgan issiqlik dvigatelining bajargan foydali ishi quyidagiga teng:

$$A_{foy} = Q_1 - Q_2$$

Q_1 – isitgichdan olingan issiqlik miqdori.

Q_2 – sovitgichga berilgan issiqlik miqdori.

Issiqlik dvigatelining foydali ish koeffitsiyenti

Issiqlik dvigatelining foydali ish koeffitsiyenti Sadi Karno tomonidan hisoblab topilgan.

$$\eta = \frac{A_{foy}}{Q_1} = \frac{Q_1 - Q_2}{Q_1} \quad \text{yoki} \quad \eta = \frac{Q_1 - Q_2}{Q_1} \cdot 100\%$$

$$\eta = \frac{T_1 - T_2}{T_1} \cdot 100\%$$

18-mashq 1-masala

Ishchi jism (gaz) isitkichdan 840 J issiqlik oldi. Agar issiqlik dvigateli foydali ish koeffitsiyenti 30% bo‘lsa, gaz qancha ish bajaradi?

Berilgan:

$Q_1 = 840$ J

$\eta = 30\%$

Topish kerak: $Af - ?$

Formula:

$\eta = \frac{Af}{Q_1} \cdot 100\%$

$Af = \frac{\eta Q_1}{100\%}$

Yechish:

$Af = \frac{30\% \cdot 840\ J}{100\%} = 252$ J

Javob: $Af = 252\ J$

Mustaqil bajarish uchun topshiriqlar

1. Karno sikli qanday jarayonlardan tashkil topgan?

2. Issiqlik mashinalari bajargan foydali ish qanday aniqlanadi?

3. Dvigatelning foydali ish koeffitsiyenti qanday hisoblanadi?

4. 91-betdagi 2 va 3- testlarni yechish.